

Network Dissection:

Quantifying Interpretability of Deep Visual Representations

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Presented by Anat Kleiman, Gustaf Ahdritz, Xin Tang, and Luke Bailey

Presentation Roadmap

- **Introduction:**

- Motivation
- Questions paper aims to answer
- Related Works

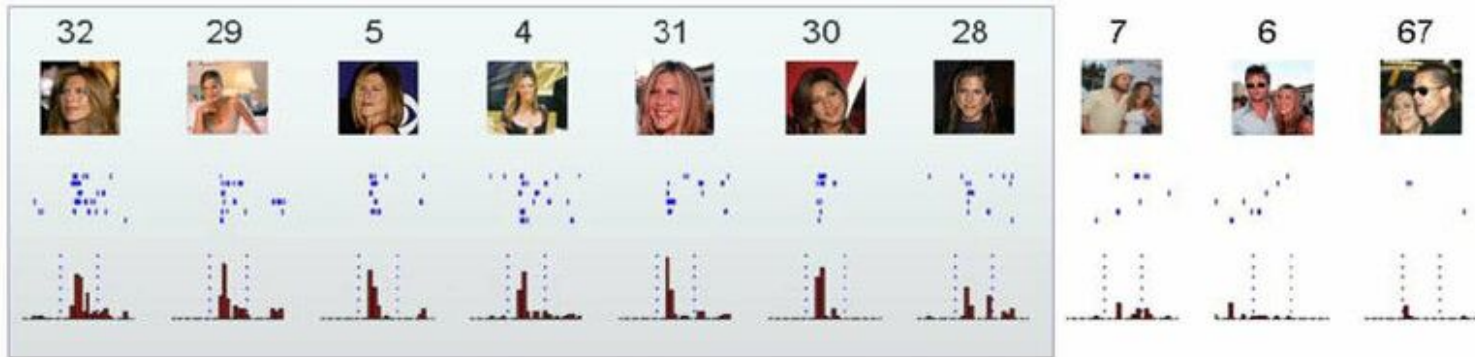
- **Method**

- **Experiments**

- Training Conditions
- Discrimination
- Layer Width

How is semantic visual concept represented in the brain?

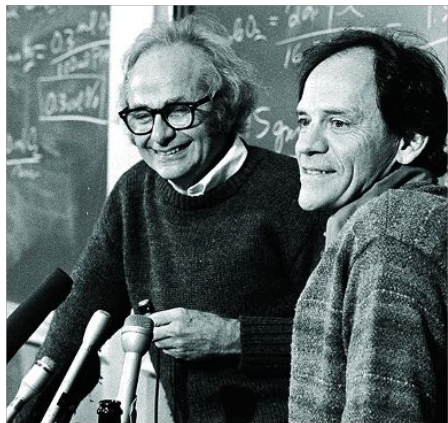




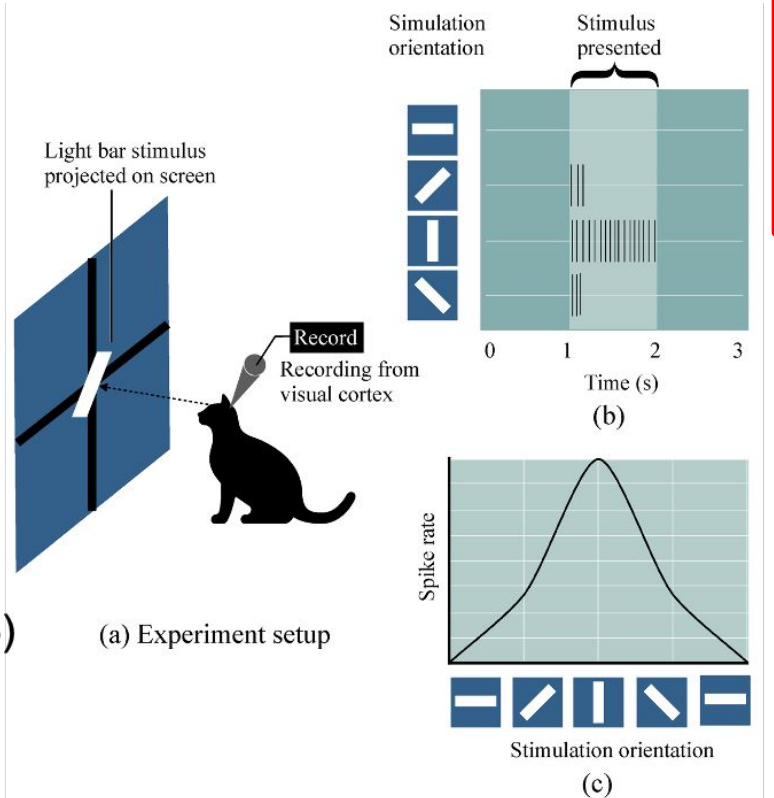
A neuron that only fires for Jennifer Aniston



Disentangled representation in visual cortex

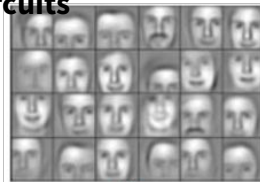


David Hubel (1926-2013)
Torsten Wiesel (1924-)



visual system and visual processing
Nobel Prize 1981

Jennifer Aniston neuron is in high-level
neural circuits



object models



object parts
(combination
of edges)



edges



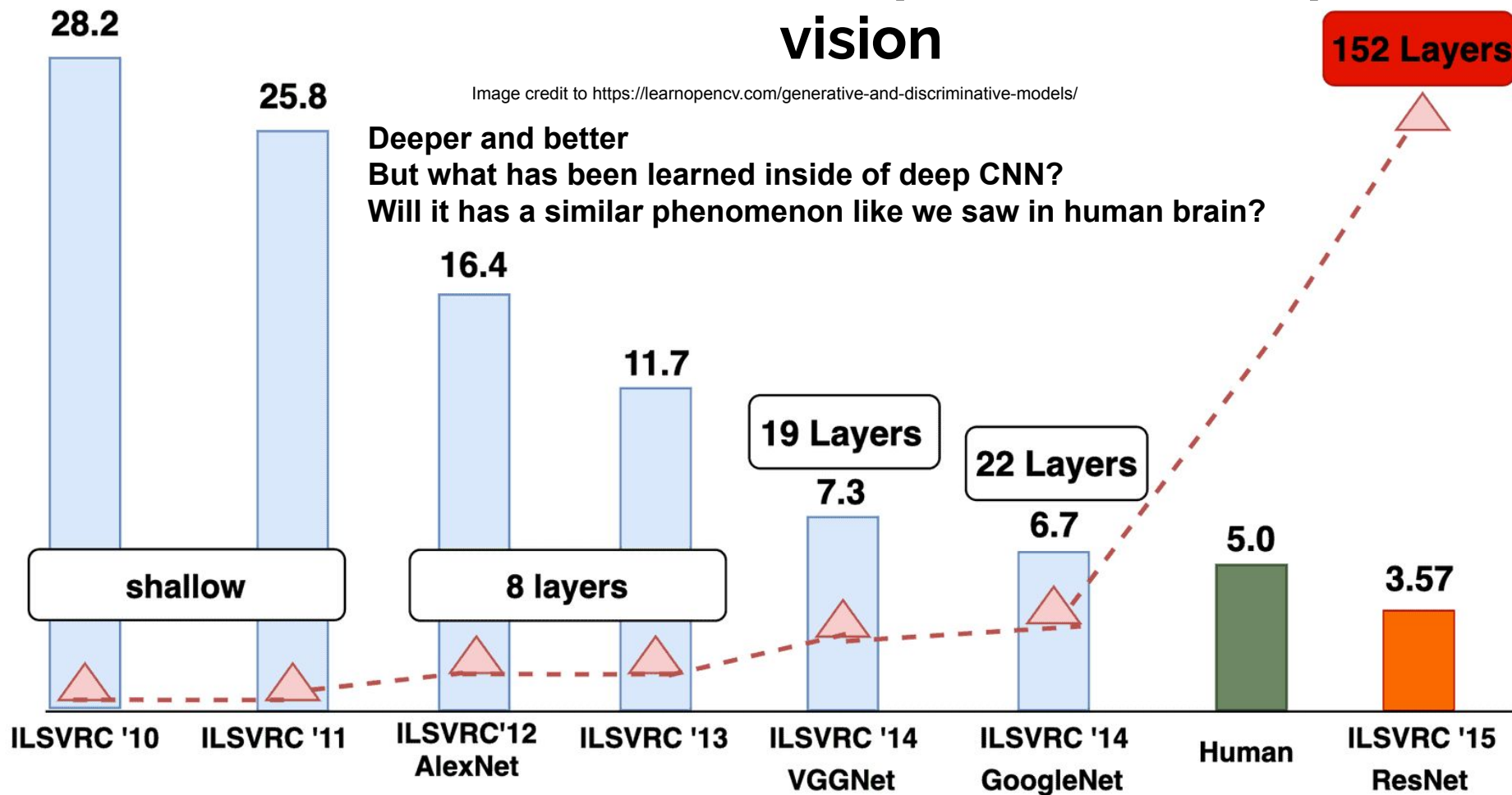
pixels

Hierarchical Coding

ImageNet Classification Top-5 Error (%)

Deep CNN for computer vision

Image credit to <https://learnopencv.com/generative-and-discriminative-models/>



Proposed questions

1. **What is a disentangled representation, and how can its factors be quantified and detected (in deep CNN)?**
2. **Do interpretable hidden units reflect a special alignment of feature space, or are interpretations a chimera?**
3. **What conditions in state-of-the-art training lead to representations with greater or lesser entanglement?**

Proposed questions and contributions

1. What is a disentangled representation, and how can its factors be quantified and detected?

- a. Proposed a metric, intersection over union score (IoU), to quantify the interpretability of each unit
- b. The alignment level between unit activated area and human-interpretable concepts

2. Do interpretable hidden units reflect a special alignment of feature space, or are interpretations a chimera?

- a. A semantic concept can be detected by many units
- b. A unit can detect many semantic concepts

3. What conditions in state-of-the-art training lead to representations with greater or lesser entanglement?

- a. Number of unique detectors, Layer depth, training iterations
- b. the angle of the images, input datasets
- c. Fine-tuning, supervised v.s. unsupervised

Related works

1. Generative Visualizations of Individual Units

- Mahendran et al., CVPR 2015
- Nguyen et al., NIPS 2016
- Simonyan et al., ICML 2014

2. Salience-based Visualizations of Individual Units

- Deconvolution: Zeiler et al., ECCV 2014

3. Visualizing Representations as a Whole

- t-SNE: Maaten et al., JMLR, 2008
- prototype autoencoder: Li et al., AAAI, 2018
- Yosinski et al., ICML, 2015 ...

Limitation of these works: qualitative analyses, cannot be used for comparison between models

Presentation Roadmap

- Introduction
- **Method**
- Experiments

Broden: Broadly and Densely Labeled Dataset

- Combination of multiple datasets with segmentation and image wide labels

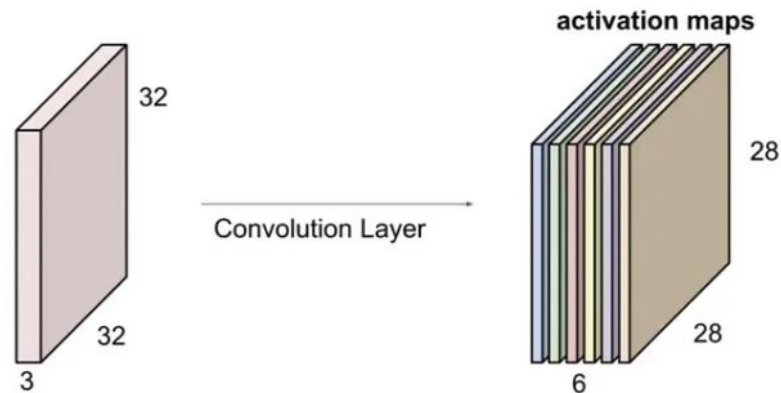
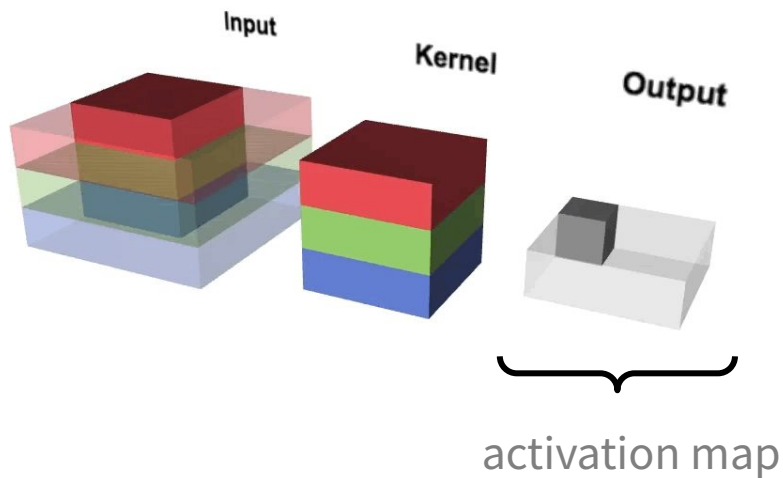


Figure 2. Samples from the **Broden** Dataset. The ground truth for each concept is a pixel-wise dense annotation.

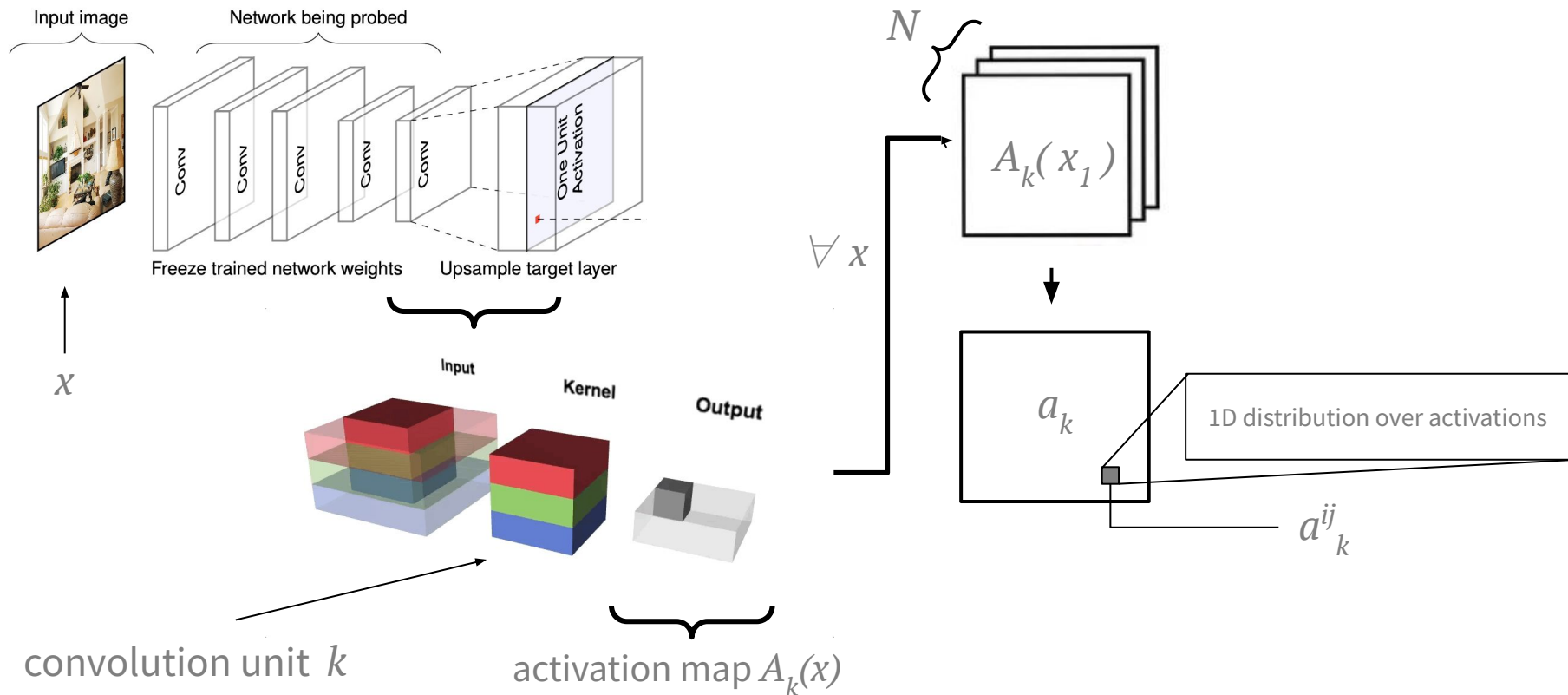
- Multiple labels can apply to the same pixel e.g. “cat, leg, black”.

Scoring Unit Interpretability

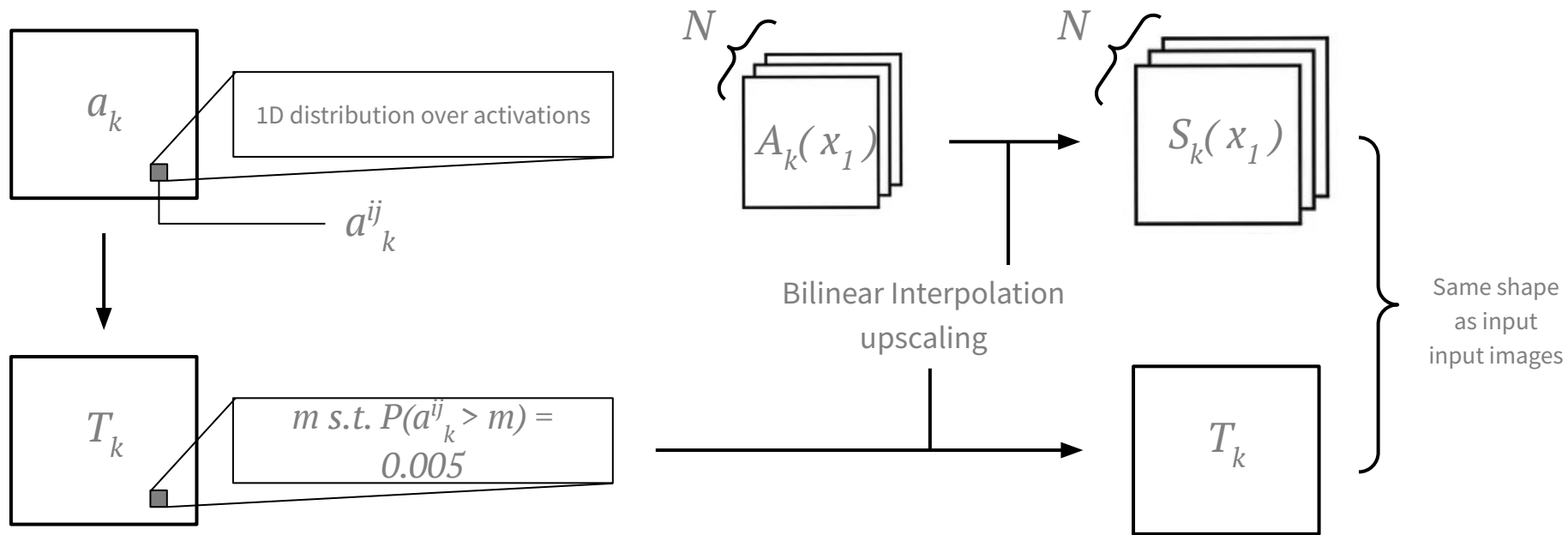
- Unit is a convolutional filter



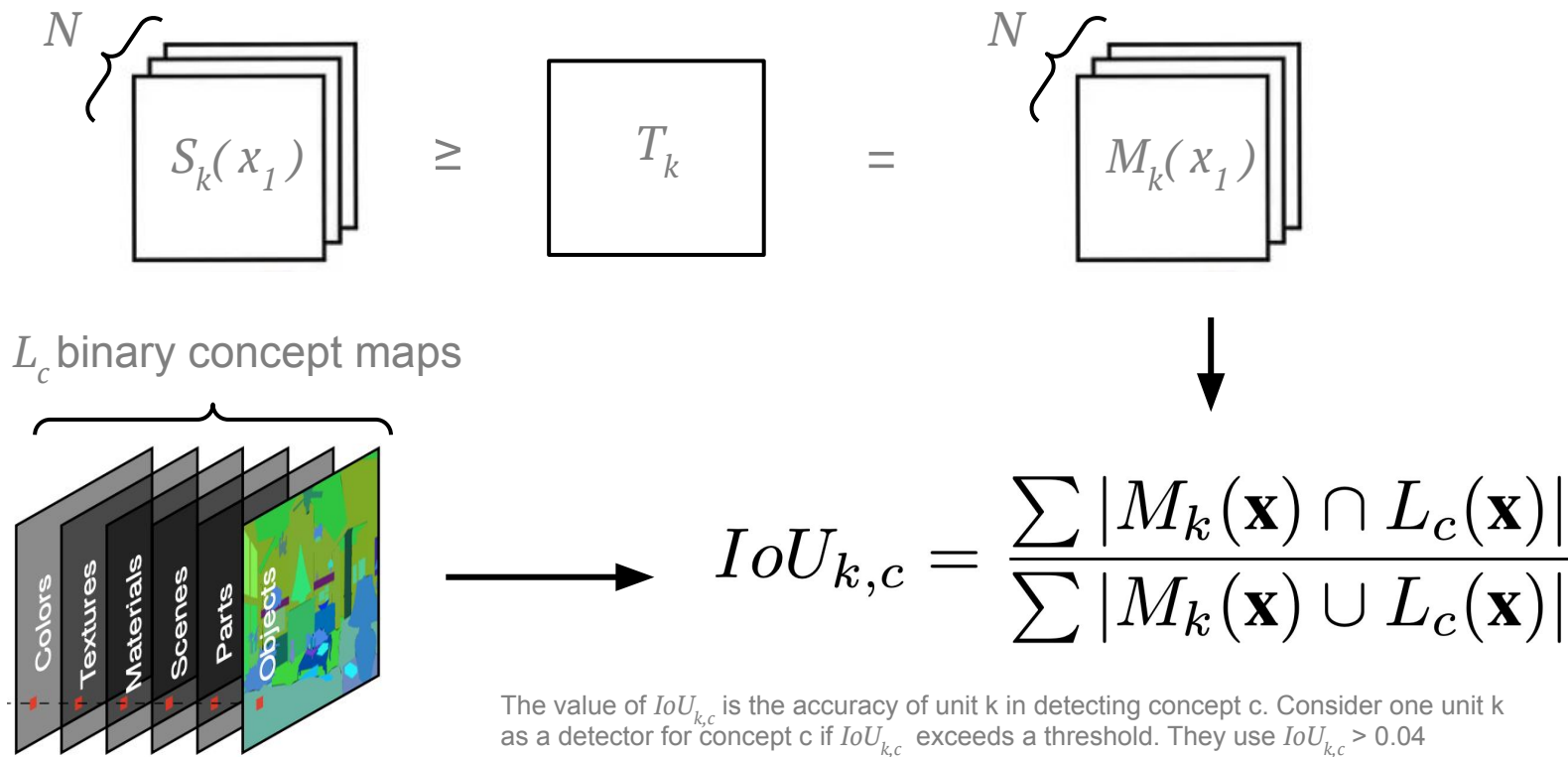
Scoring Unit Interpretability



Scoring Unit Interpretability



Scoring Unit Interpretability



Scoring Unit Interpretability

$$IoU_{k,c} = \frac{\sum |M_k(\mathbf{x}) \cap L_c(\mathbf{x})|}{\sum |M_k(\mathbf{x}) \cup L_c(\mathbf{x})|}$$

- Changing *IoU* threshold changes number of concept detectors but not orderings between networks.
- One unit might be detector for multiple concepts, they just choose the top ranked concept for an individual unit.
- Interpretability of a layer is the number of unique concepts aligned with units

Presentation Roadmap

- Introduction
- Method
- Experiments

Experiments

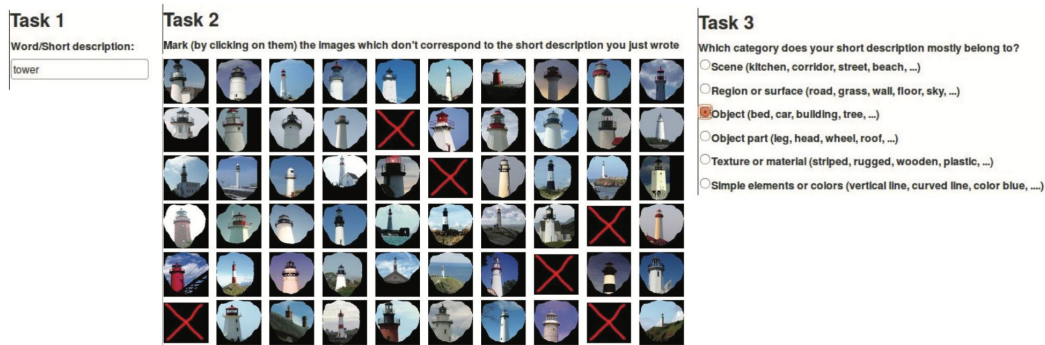
Scene-centric data set with categories such as kitchen, living room, and coast

Table 2. Tested CNNs Models

Training	Network	Data set or task
none	AlexNet	random
Supervised	AlexNet	ImageNet, Places205, Places365, Hybrid.
	GoogLeNet	ImageNet, Places205, Places365.
	VGG-16	ImageNet, Places205, Places365, Hybrid.
	ResNet-152	ImageNet, Places365.
Self	AlexNet	context, puzzle, egomotion, tracking, moving, videoorder, audio, crosschannel,colorization. objectcentric.

Experiment 1: Human Evaluation of Interpretations

1. **Identify Interpretable units** - units that raters agreed with ground-truth interpretations from [*]
2. Raters shown 15 images with highlighted patches showing the most highly-activating regions for each unit in AlexNet trained on Places205, and asked to decide (yes/no) whether a given phrase describes most of the image patches.
3. **Find network dissection** - the portion of interpretations generated by method that were rated as descriptive
4. **Human Consistency** - portion of ground-truth labels that were found to be descriptive by a second group of raters



[*] B. Zhou, A. Khosla, A. Lapedriza, A. Oliva, and A. Torralba. Object detectors emerge in deep scene cnns. International Conference on Learning Representations, 2015.

Figure 6: AMT interface for unit concept annotation. There are three tasks in each annotation.

Experiment 1: Human Evaluation of Interpretations

Table 3. Human evaluation of our Network Dissection approach. Interpretable units are those where raters agreed with ground-truth interpretations. Within this set we report the portion of interpretations assigned by our method that were rated as descriptive. Human consistency is based on a second evaluation of ground-truth labels.

	conv1	conv2	conv3	conv4	conv5
Interpretable units	57/96	126/256	247/384	258/384	194/256
Human consistency	82%	76%	83%	82%	91%
Network Dissection	37%	56%	54%	59%	71%

Experiment 2: Axis-Aligned Interpretability

Two hypotheses:

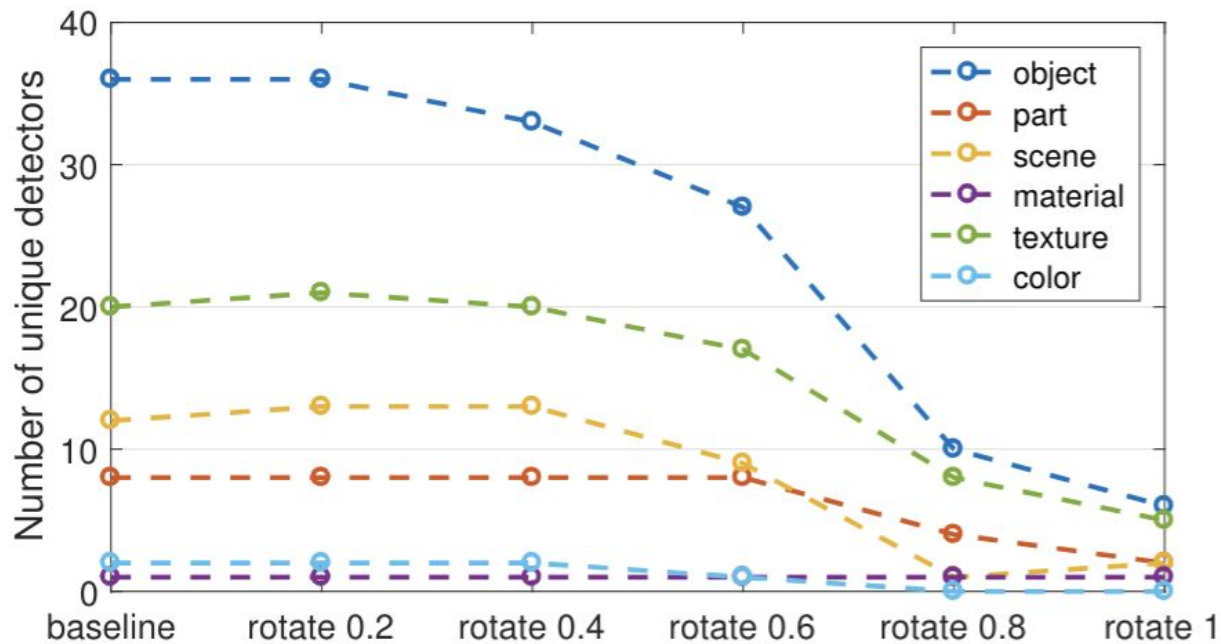
1) Concepts appear in every direction

- **Default hypothesis.** Single units not much more interpretable than combinations of units.

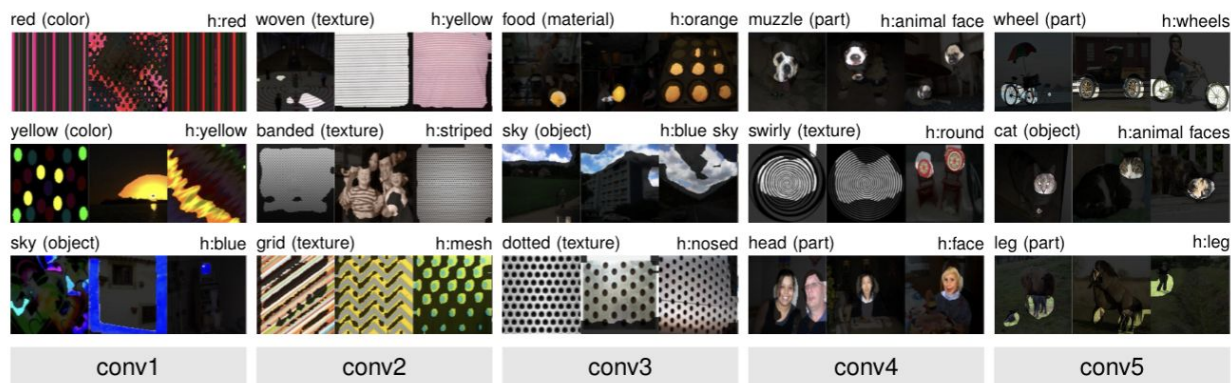
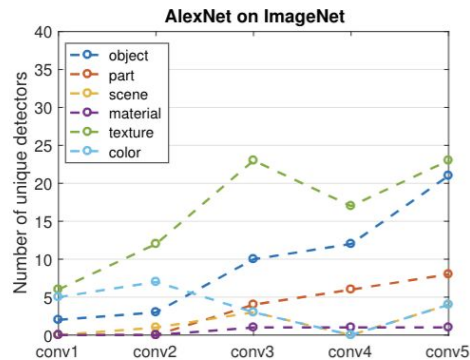
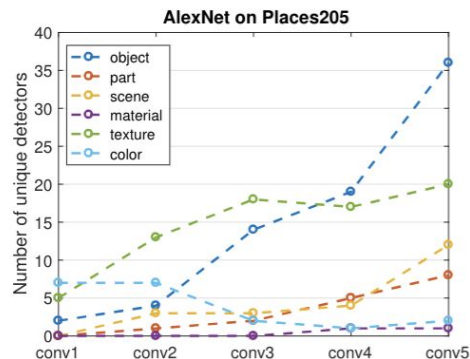
2) Concepts are rare + the model converges to a special, semantically rich basis

- The model's natural basis is a meaningful decomposition.

Experiment 2: Axis-Aligned Interpretability



Experiment 3: Concepts by layer



Experiment 4: Network Architectures

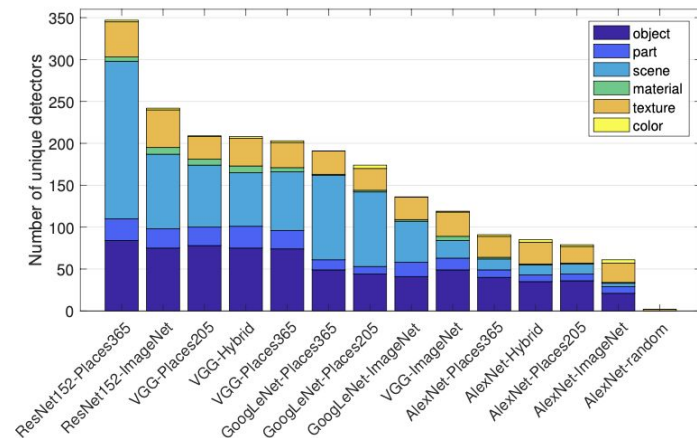


Figure 7. Interpretability across different architectures and training.

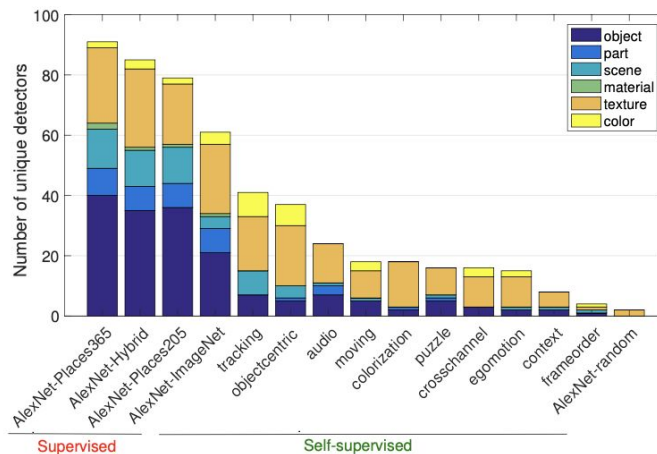


Figure 8. Semantic detectors emerge across different supervision of the primary training task. All these models use the AlexNet architecture and are tested at conv5.

Experiment 5: Training Conditions

Varied training conditions:

1) Weight initializations

- **Minimal Effect:** Models converge to similar levels of interpretability

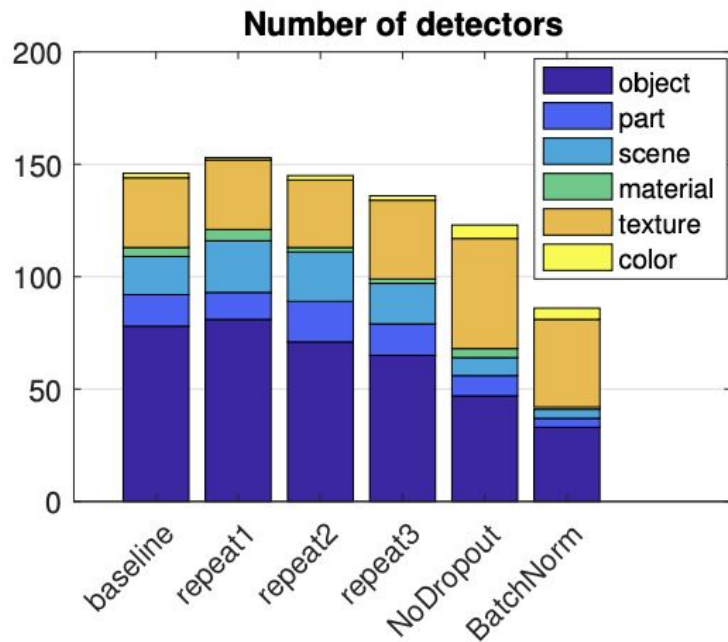
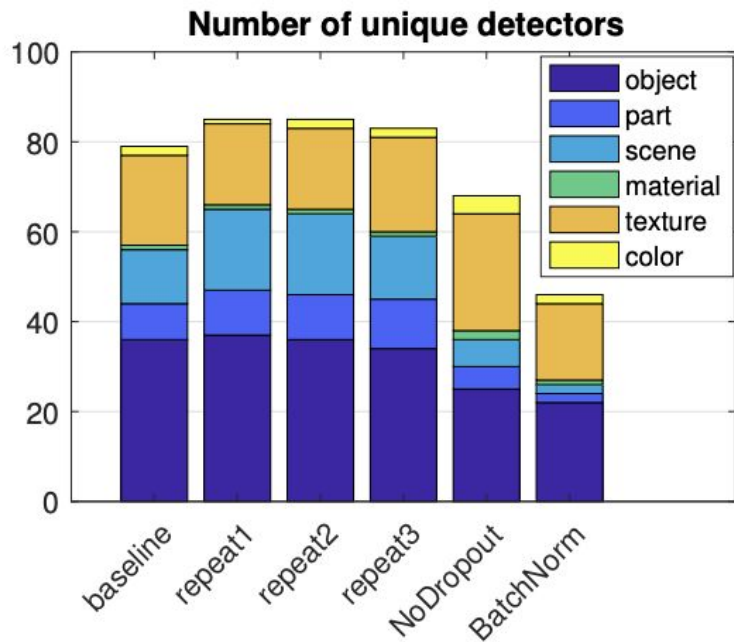
2) Dropout

- **Some Effect:** Lack of dropout leads to more “texture” and less “object” detectors,

3) Batch Normalization

- **Significant Effect:** Interpretability decreased significantly

Experiment 5: Training Conditions

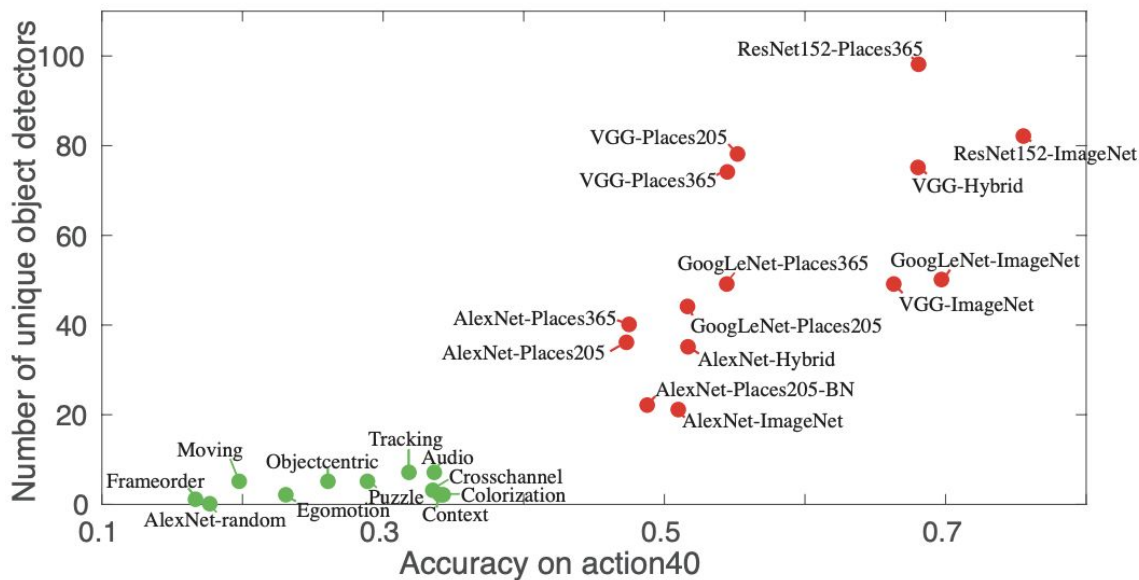


Experiment 6: Discrimination

Benchmark high-level activations on a new task:

- Across several Deep NNs, extract activation from high CNN layers
- Train a linear SVM on a new *action recognition task*
- Compute classification accuracy

Experiment 6 Discrimination



Result: Positive correlation between object detectors and classification accuracy → Encouraging **concept detection** can improve **discrimination**

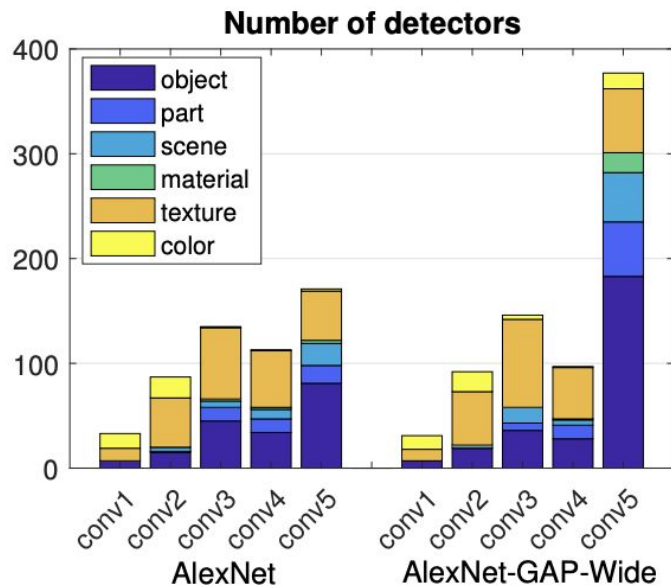
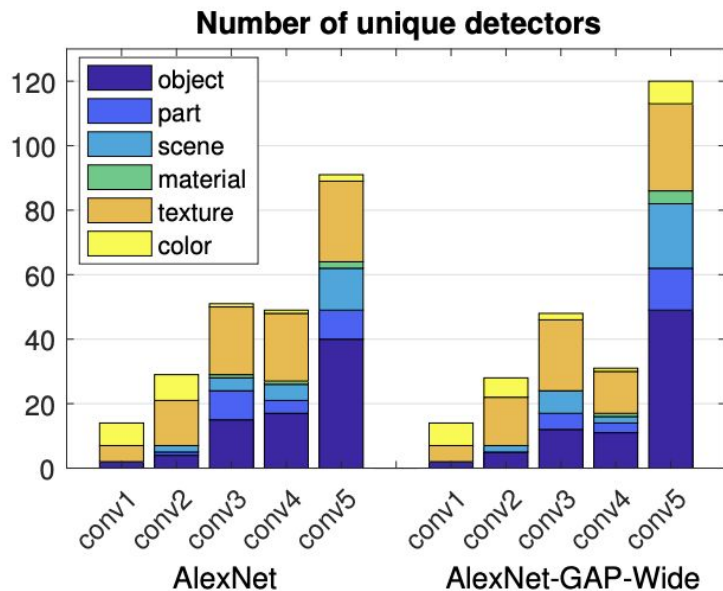
Experiment 7: Width

Effect of layer width (number of units in a layer):

Increased layer width retains similar accuracy, but much more **concept detectors**

- # Detectors increased both at increased layer and in network generally
- Increase has a threshold

Experiment 7: Width



Discussion Questions

- 1) **Distribution Understanding:** Concept detectors from a particular dataset betray something about the underlying distribution. How do you think this can be applied in the real world (e.g. bias detection)?
- 2) **Single unit to circuit:** This method interprets single units, could it be extended to larger circuits and would this be useful?
- 3) **Beyond vision:** This method is deeply tied to the vision domain. Could it be extended to other domains such as natural language?

Interpretability Beyond Feature Attribution: Quantitative Testing with Concept Activation Vectors (TCAV)



Kim et al. (2018)

Presented by Lucia Gordon, Matthew Nazari, Catherine Yeh

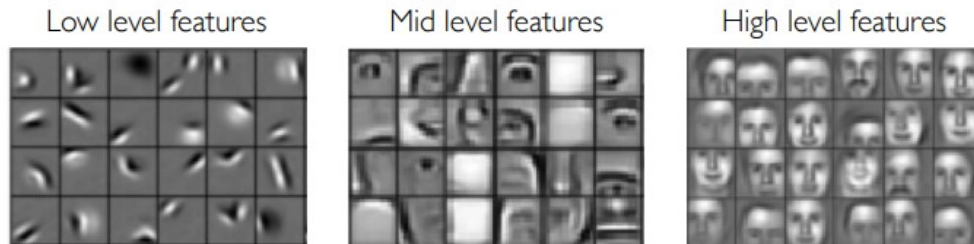


Thoughts on **Concept-Based** Explanations So Far?



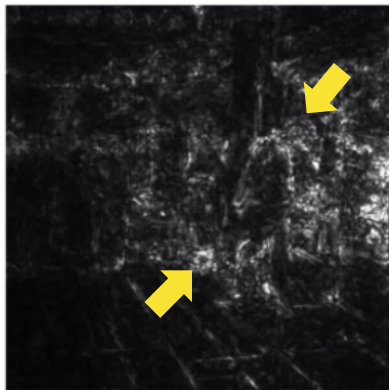
Motivation + Problem Statement

- Interpreting deep learning models is crucial to understanding their behavior, ensuring accurate predictions, and reflecting our values
- But remains a big challenge due to size, complexity, and opacity of ML models
- Additionally, many systems operate on **low-level features** (e.g., pixel values) rather than **high-level concepts** (e.g., face) that are human-interpretable



Motivation + Problem Statement

prediction:
Cash machine



Were there more pixels on the cash machine than on the person?

Did the 'human' concept matter?
Did the 'wheels' concept matter?

Which concept mattered more?

Is this true for all other cash machine predictions?

Problem:

- We can't express these concepts as pixels
- And they weren't our input features

Image from [Been Kim](#)

Summary of Contributions

- Introduce **Concept Activation Vectors (CAVs)**: way to interpret a neural network's internal state in terms of human-friendly concepts
- Key idea is to use the high-dimensional internal state of a neural net as an aid not an obstacle
- Main contribution is a new linear interpretability method, **Testing with CAV (TCAV)**, that quantifies model sensitivity to a high-level concept learned by a CAV for a particular class

E.g., "striped"

a



E.g., "zebras"

b



Goals of TCAV

- **Accessibility:** requires little to no ML expertise
- **Customization:** adaptable to any concept, even outside of training
- **Plug-in readiness:** works without retraining/modifying ML models
- **Global quantification:** can interpret entire classes with a single quantitative measure



Assessed with experiments + human evaluation

Related Work

Interpretability methods:

(Goodman & Flaxman, 2016)

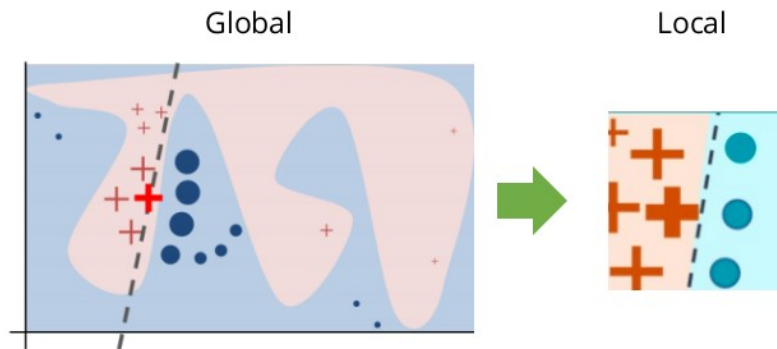
- *Inherently interpretable* models vs. *post-hoc* explanations

(Kim et al., 2014; Doshi-Velez et al., 2015)

- **Perturbation-based methods:** e.g., LIME/SHAP

○ *local* vs. *global*

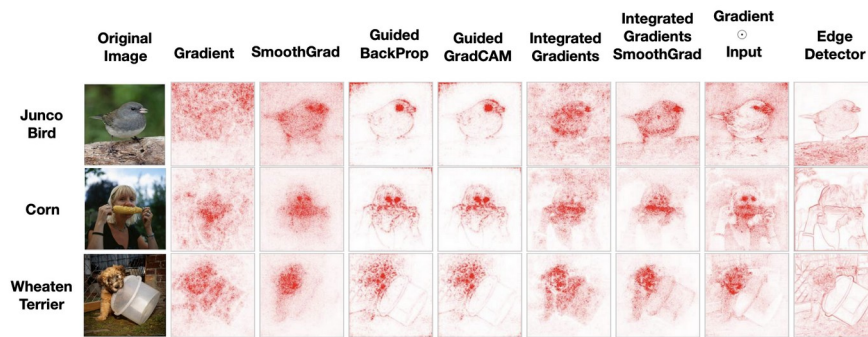
(Ribeiro et al, 2016; Lundberg & Lee, 2017)



Related Work

Interpretability methods in neural networks

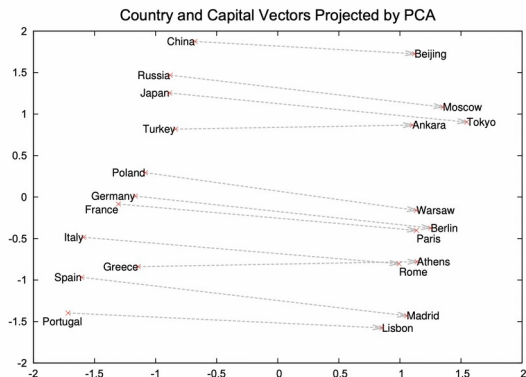
- Limitations w/ **saliency methods**
 - Local explanation (Erhan et al., 2009;
 - Lack customization (Smilkov et al., 2017)
 - Vulnerable to adversarial attacks (Ghorbani et al., 2017)
 - Insensitivity to randomization (Adebayo et al., 2018)



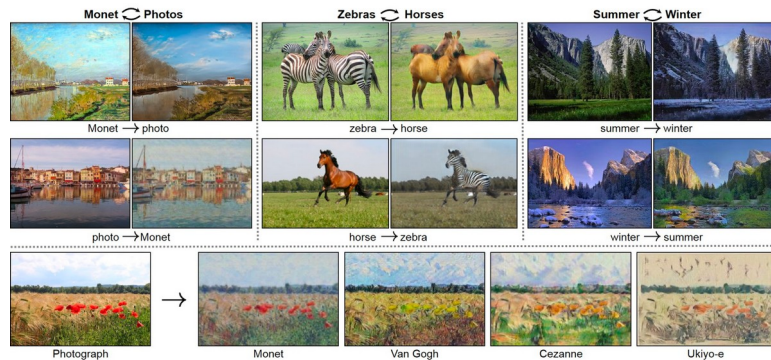
Related Work

Linearity in neural network + latent dimensions

- Meaningful information can be learned from simple *linear* classifiers
(Bau et al., 2017; Alain & Bengio. 2016)
- Mapping *latent* dimensions to human *concepts*
(Mikolov et al., 2013; Zhu et al., 2017)



word2ve
c



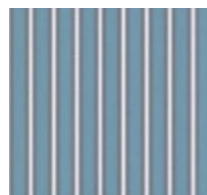
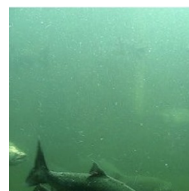
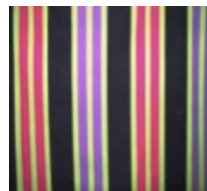
cycleGA
N

Approach: Defining the CAV

Consider the fully connected layer $f_l : \mathbb{R}^n \rightarrow \mathbb{R}^m$
and the concept of interest

Collect a set of examples P_C of that concept and a
negative set of examples N that don't

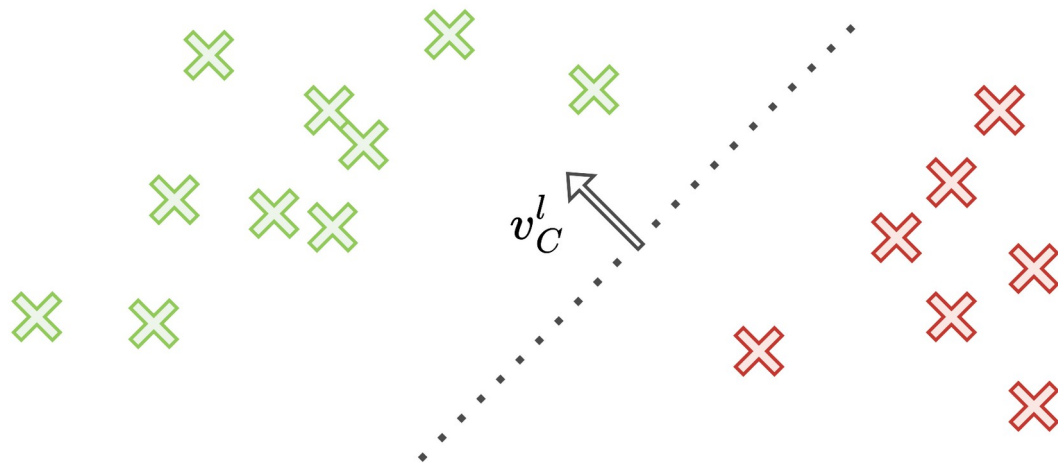
Define the CAV to be a vector orthogonal to a decision
boundary between activation $\{f_l(\mathbf{x}) : \mathbf{x} \in P_C\}$ and
 $\{f_l(\mathbf{x}) : \mathbf{x} \in N\}$



Stripes P_C

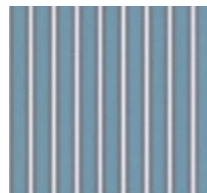
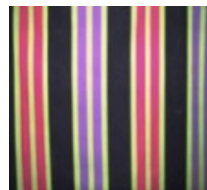
Not N
stripes

Approach: Visualizing the CAV



$$\text{X} \in \{f_l(x) : x \in P_C\}$$

$$\text{X} \in \{f_l(x) : x \in N\}$$



Stripes P_C



Not stripes N

Approach: Gauging “concept sensitivity”

Saliency maps gauge sensitivity of prediction $h_k(\mathbf{x})$ with respect to per-pixel perturbations

$$S_{C,k,l}(\mathbf{x}) = \nabla h_{l,k}(f_l(\mathbf{x})) \cdot \mathbf{v}_C^l$$

With CAV, we can gauge sensitivity of predictions $h_{l,k}(f_l(\mathbf{x}))$ **towards a concept** $\mathbf{v}_C^l$ at an entire layer

Approach: Testing with CAV (TCAV)

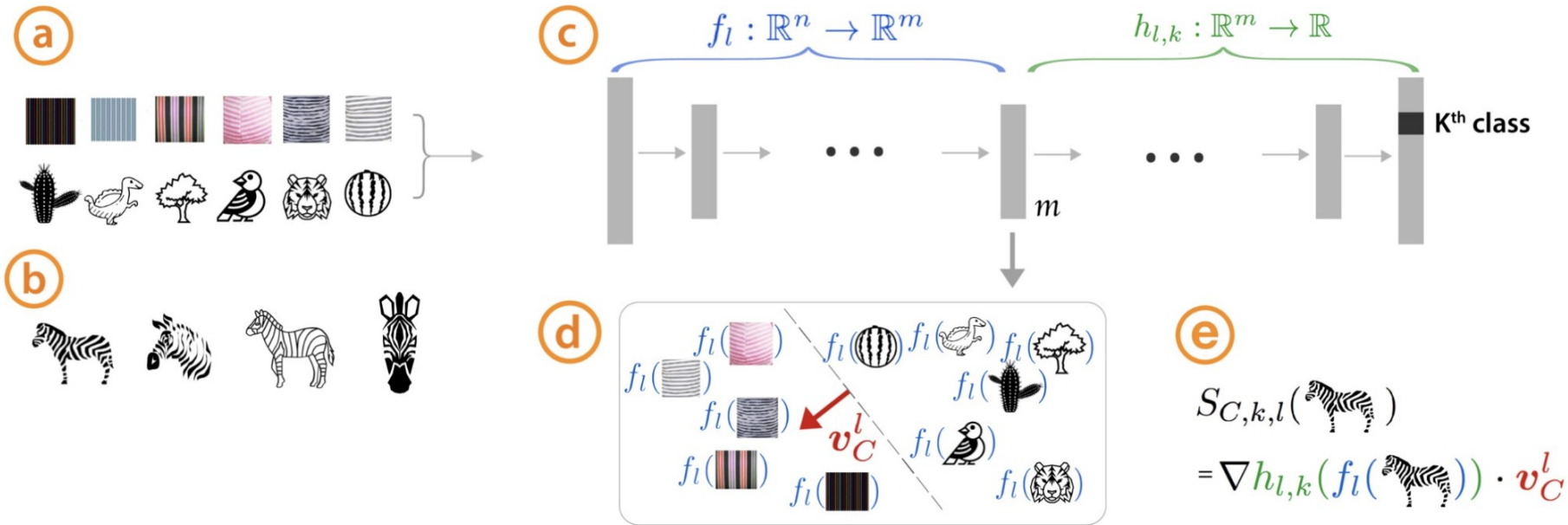
$$\text{TCAV}_{Q_{C,k,l}} = \frac{|\{\mathbf{x} \in X_k : S_{C,k,l}(\mathbf{x}) > 0\}|}{|X_k|}$$

$\text{TCAV}_{Q_{C,k,l}}$ measures the fraction of inputs whose activations were influenced by a concept

This provides interpretation global to a particular class

A t -test can safeguard against meaningless CAVs

Approach Summary



Results: Sorting Images with CAVs

Class:
Stripes
Concept:
CEO

CEO concept: most similar striped images



CEO concept: least similar striped images



Class: necktie
Concept: model woman

Model Women concept: most similar necktie images



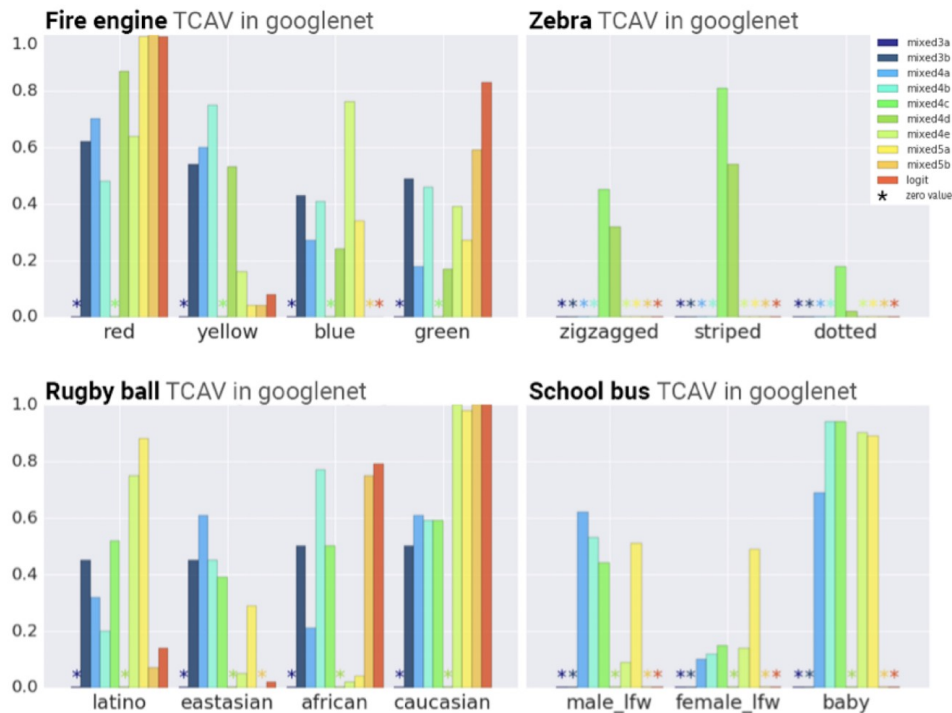
Model Women concept: least similar necktie images



- Confirmation that the CAVs correctly reflect the concept of interest
- Sorting procedure can reveal biases used to learn the CAV

Results: Gaining Insights with TCAV

- x-axis: CAV
- y-axis: TCAV score = conceptual sensitivity of classification to concept
- Each bar is a different layer of the NN with a statistically significant CAV
- Layers closer to logit layer have a greater influence on the prediction

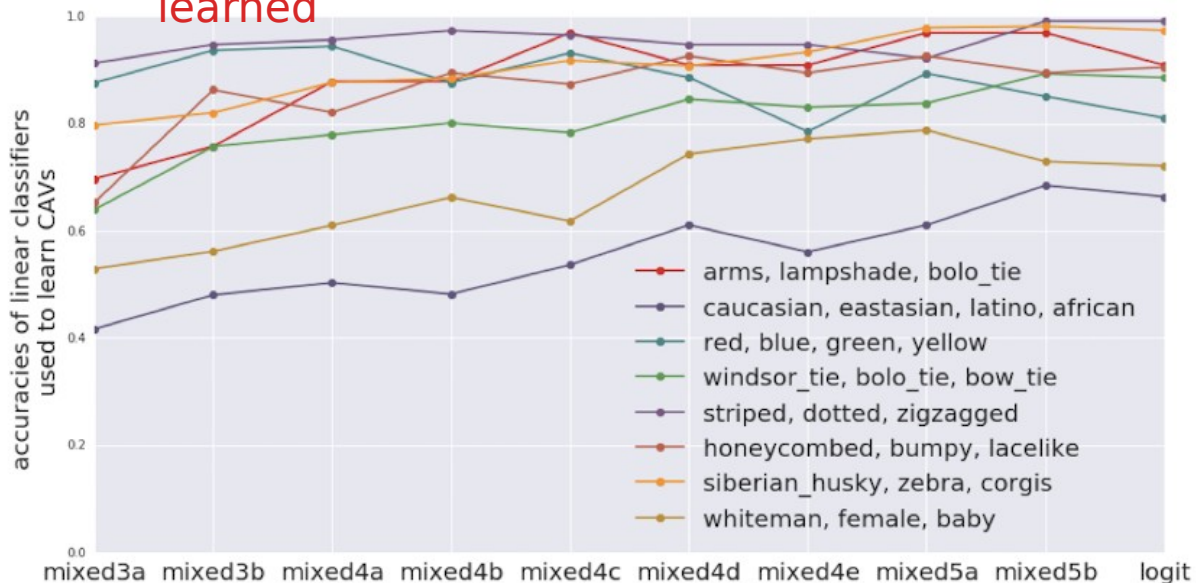


- Matches intuition
 - “Red” is important for “fire engine”
 - “Striped” is important for “zebra”
- Biases
 - “Caucasian” is important for “rugby ball”
- Statistical significance test successfully removed spurious CAVS
 - “Dotted” is not important for “zebra”
- Quantitatively confirm qualitative

Results: TCAV for Where Concepts are Learned

- Simple concepts (ex. colors, patterns) reach a high accuracy at low layers
- Complex concepts (ex. age, sex, objects) don't reach high accuracy until higher layers
- Confirming past findings that lower layers = feature detectors and higher layers = classifiers

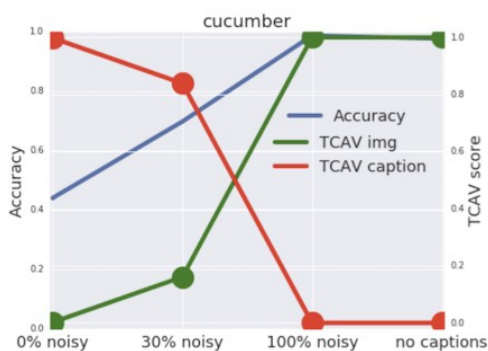
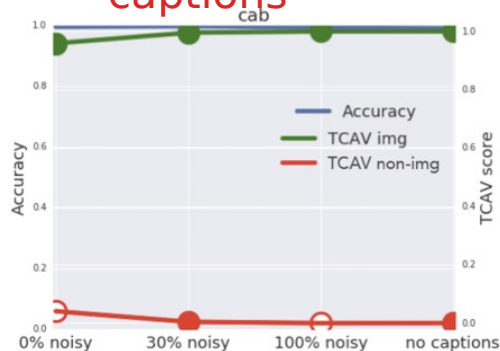
Accuracies of different CAVs at different layers, showing where each concept is learned



Results: Controlled Experiment with Ground Truth



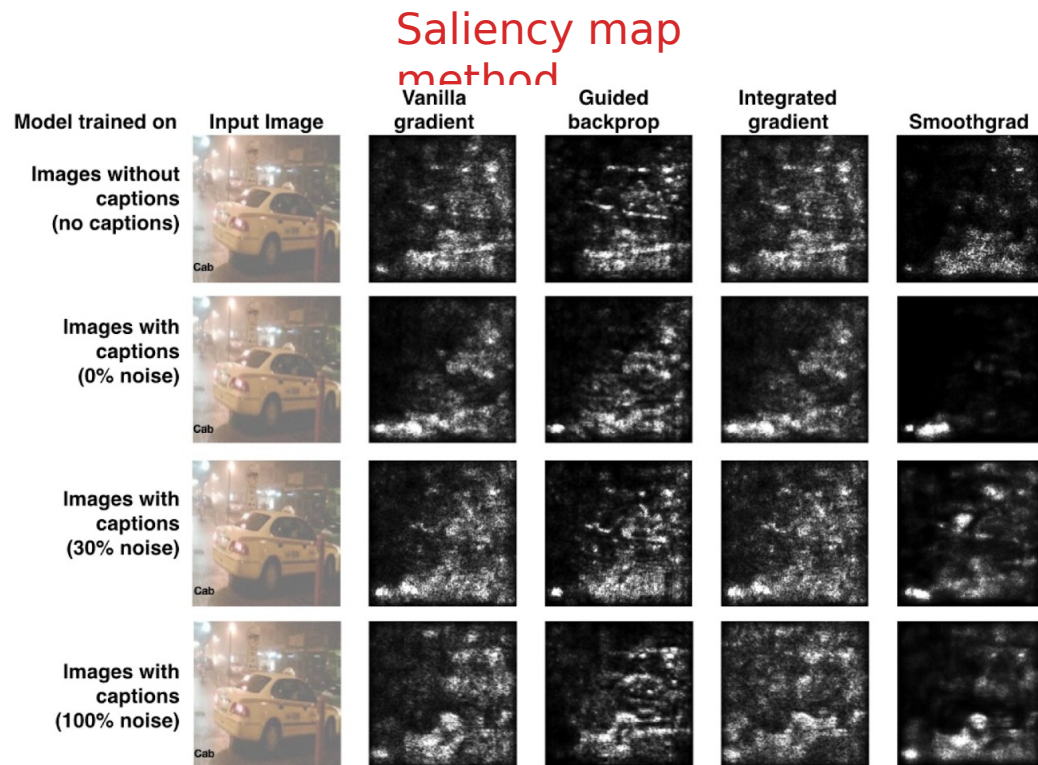
- TCAV img → image concept CAV
- TCAV non-img → caption concept CAV
- Accuracy → tested on images without captions



- Cab: image concept more important than caption concept regardless of noise
- Cucumber: caption concept more important when caption is likely to appear
- TCAV reflects ground-truth
 - Only image is important → high accuracy
 - Only caption is important → low accuracy

Results: Evaluation of Saliency Maps

We know that for “cab” the “image” concept is most important, but this is not reflected in saliency maps → superiority of TCAV interpretability method

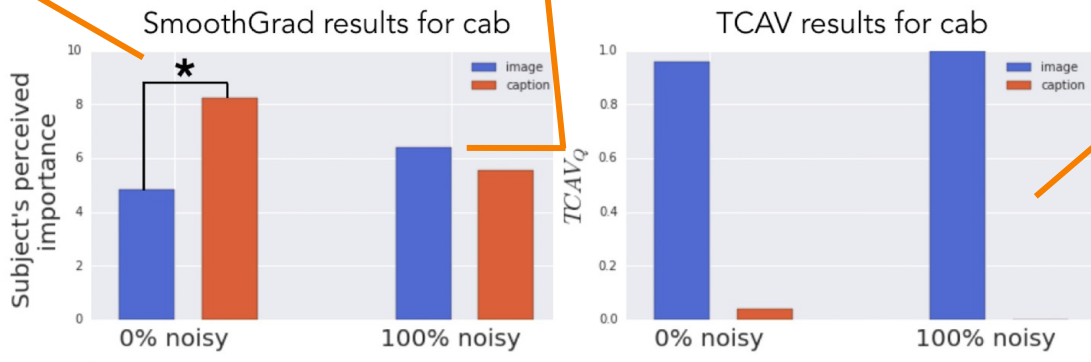


Results: Evaluation of Saliency Maps with Human Subjects

Subjects incorrectly thought the caption was more important than the image

Subjects could not discern a difference in importance between the image and caption

TCAV score correctly reflects which concept is most important



Saliency maps are misleading!

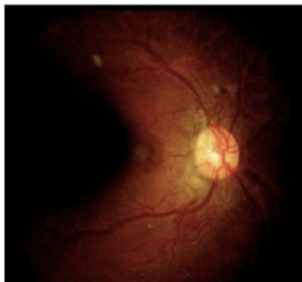
Results: TCAV for a Medical Application

Green = relevant concept

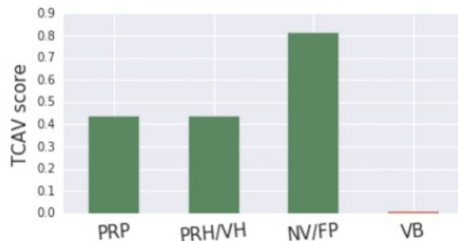
irrelevant concept

Red =

DR level 4 Retina



TCAV for DR level 4



TCAV score shows model successfully distinguishes relevant and irrelevant concepts for level 4 diagnosis

DR level 1 Retina

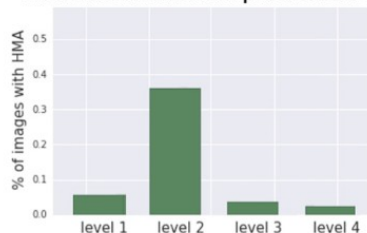


TCAV for DR level 1



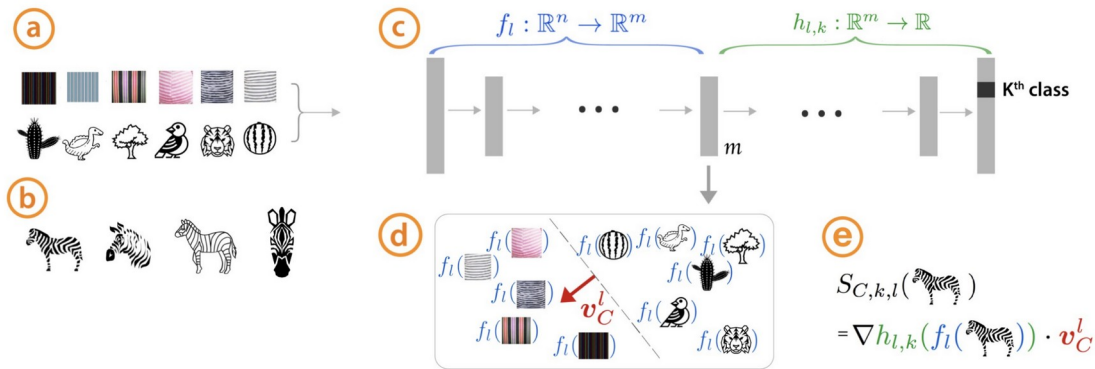
TCAV score shows model is giving too much importance to HMA concept for level 1 diagnosis; HMA is relevant to level 2 not level 1 → use this to debug model

HMA distribution on predicted DR



Conclusions

- **Roadmap:** Gradient → Attention → Concept based approaches (TCAV!)
- **Limitations**
 - Evaluated only on computer vision tasks
 - Statistical significance testing — is this rigorous?



Discussion Questions

- How does this method compare to e.g., saliency maps?
- How would you adapt TCAV for other types of data (e.g., audio/video) and what would that look like?
- Thoughts on TCAV for adversarial example identification (Appendix A)?
- Do you think there could be adversarial images that allow meaningless CAVs to pass the statistical significance test?

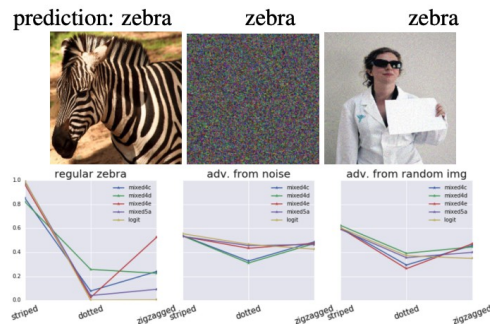


Figure 11. Two types of adversarial images that are classified as zebra. In both cases, the distribution of TCAVQare is different from that of a normal zebra.